Lesson 1: Similar Polygons and Scale Factor

Assignment

1. Each small square is 1 unit. For each copy, is it similar to the original? If yes, what number was every length multiplied by?



A: _____ B: _____ C: _____ D: _____

2. True or false? Explain each answer.

(a) Two congruent figures are also similar.

(b) When a figure is enlarged, its angles get bigger.

(c) A scale factor of 0.5 makes an enlargement.

3. $PQRS \sim WXYZ$.

(a) Complete the table of corresponding parts.

In $PQRS$	$\angle P$	$\angle Q$	$\angle R$	$\angle S$
In $WXYZ$				

In $PQRS$	PQ	QR	RS	SP
In $WXYZ$				

(b) Complete the equal ratios: $\dfrac{WX}{PQ} = \dfrac{\quad}{QR} = \dfrac{YZ}{\quad} = \dfrac{\quad}{\quad}$

4. $\triangle DEF \sim \triangle KLM$. $\angle D = 38°$ and $\angle E = 97°$. Find $\angle F$, $\angle K$, $\angle L$ and $\angle M$.

5. Pentagon $ABCDE \sim$ pentagon $VWXYZ$. $\angle V = 108°$, $\angle W = 120°$, $\angle X = 95°$ and $\angle Y = 110°$. You don't need a picture: use the naming order.

(a) Find $\angle Z$. (The angles in a pentagon add up to 540°.)

(b) Find the five angles of $ABCDE$.

6. Complete the table.

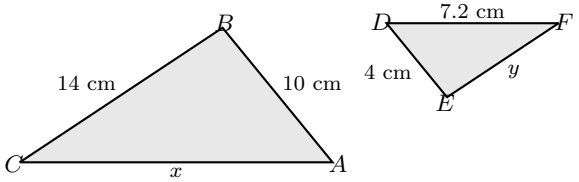
	Original	**New**	**Scale factor k**	**Type**
(a)	5 cm	20 cm		
(b)	20 m	12 m		
(c)	3.2 in	11.2 in		
(d)	9 m	9 m		
(e)	50 cm	10 cm		

7. Change to the same units first, then find the scale factor.

(a) A cookie cutter is 6 cm wide. A sign shop makes a giant cookie-shaped sign for a bakery in Airdrie that is 1.5 m wide. Find the scale factor from the cookie cutter to the sign.

(b) A real canoe is 4.8 m long. A toy canoe is 16 cm long. Find the scale factor from the real canoe to the toy, as a fraction with 1 on top.

8. $\triangle ABC \sim \triangle DEF$. The diagrams are not drawn to scale.



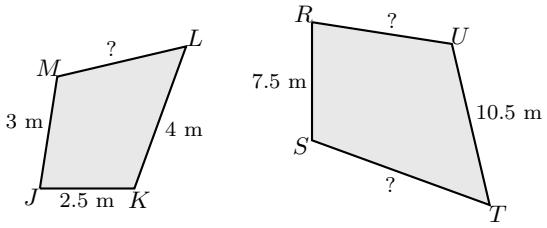
(a) Find the scale factor from $\triangle ABC$ to $\triangle DEF$. Is it an enlargement or a reduction?

(b) Find y (side EF).

(c) Find x (side CA).

9. $JKLM \sim RSTU$. The diagrams are not drawn to scale.

(a) Find the scale factor from $JKLM$ to $RSTU$.



(b) Find ST and UR.

(c) Find LM.

10. Ravi takes a photo on his phone that is 12 cm wide and 9 cm tall. In class, it is shown on a projector screen as a similar picture that is 1.8 m wide.

(a) Find the scale factor. (Change 1.8 m to centimetres first.)

(b) How tall is the picture on the screen?

(c) The screen is 1.2 m tall. Does the whole picture fit? Explain.

11. Kayla collects $1:64$ model cars.

(a) Her favourite real car is 4.48 m long. How long is the model, in centimetres?

(b) The model is 2.25 cm tall. How tall is the real car, in metres?

12. A team pennant is a triangle with sides 30 cm, 30 cm and 12 cm. The team orders a similar giant pennant for the arena. Its short side is 54 cm.

(a) Find the scale factor and the lengths of the two long sides of the giant pennant.

(b) Gold trim goes all the way around the giant pennant. The trim costs \$2.49 per metre. Find the cost of the trim before tax.

13. Jaxon is reducing a drawing. A 20 cm side becomes 8 cm. Here is his work for the new length of a 15 cm side:

$k = 20 \div 8 = 2.5$ new side $= 15 \times 2.5 = 37.5$ cm

Find his mistake and give the correct answer. How could he have seen that his answer was wrong?

14. A small triangle has sides 5 cm, 7 cm and 9 cm. It is enlarged with a scale factor of 3.

(a) Find the sides of the large triangle.

(b) Find both perimeters. How does the large perimeter compare to the small one? Explain why.

15. Which scale factor makes a **reduction**?

A. 1.25 **B.** 0.8 **C.** 1 **D.** 2

16. $ABCD \sim WXYZ$. Which side corresponds to CD?

A. WX **B.** XY **C.** YZ **D.** ZW

17. Numerical Response A $1:150$ model of a ferry is 42 cm long. The length of the real ferry, in metres, is _____ m.

18. Numerical Response A triangle has sides 4.5 cm, 6 cm and 7.5 cm. It is enlarged with a scale factor of 2.4. The longest side of the new triangle is _____ cm.

Lesson 2: Are They Similar? Checking Angles and Side Ratios

Assignment

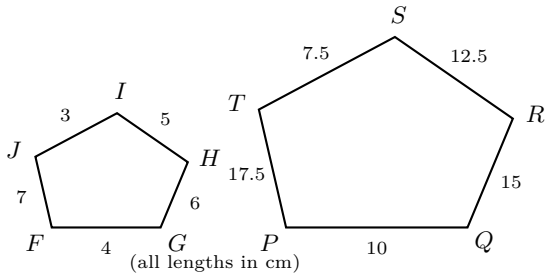
1. $ABCD \sim WXYZ$. Fill in the blanks.

(a) $\angle A$ matches _____ $\angle C$ matches _____ side AB matches _____ side CD matches _____

(b) Two polygons are similar when their corresponding _____ are all equal **and** their corresponding sides all have the same _____.

(c) To show two polygons are **not** similar, how many pairs of sides or angles must fail? _____

2. The corresponding angles of these pentagons are equal. Complete the table to check the sides. (Not drawn to scale.)



Pair	$PQRST \div FGHIJ$	**Ratio**
PQ and FG		
QR and GH		
RS and HI		
ST and IJ		
TP and JF		

Similar? _____ Scale factor: _____ Enlargement or reduction? _____

3. Quadrilaterals $ABCD$ and $EFGH$ have equal corresponding angles. The sides are: $AB = 8$ cm, $BC = 12$ cm, $CD = 10$ cm, $DA = 6$ cm and $EF = 6$ cm, $FG = 9$ cm, $GH = 7.5$ cm, $HE = 5$ cm.

(a) Find the ratio for each pair of corresponding sides (new ÷ original, with $ABCD$ as the original).

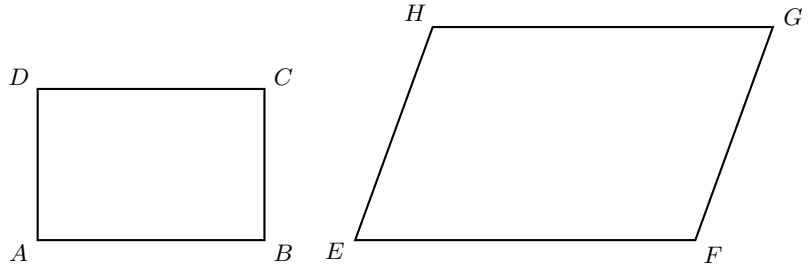
(b) Are the quadrilaterals similar? Explain.

4. Quadrilateral $KLMN$ has $\angle K = 70°$, $\angle L = 110°$, $\angle M = 95°$. Quadrilateral $STUV$ has $\angle S = 70°$, $\angle T = 110°$, $\angle V = 90°$. The vertices match in order ($K \leftrightarrow S$, ...).

(a) Find $\angle N$ and $\angle U$.

(b) Can $KLMN$ and $STUV$ be similar? Explain without knowing the sides.

5. Rectangle $ABCD$ and parallelogram $EFGH$ are drawn full size. Use a ruler and a protractor.



(a) $AB =$ _____ $BC =$ _____ $EF =$ _____ $FG =$ _____

(b) $\angle A =$ _____ $\angle E =$ _____ $\angle F =$ _____

(c) Find the side ratios. Which test passes and which test fails? Are the shapes similar?

6. Which rectangles are similar to a 6 cm by 8 cm rectangle? ($8 \div 6 \approx 1.33$)

Rectangle	**Long ÷ short**	**Similar?**
(a) 9 cm by 12 cm		
(b) 12 cm by 14 cm		
(c) 20 cm by 15 cm		
(d) 3 cm by 4 cm		
(e) 8 cm by 10 cm		

7. Most TV screens are made in a 16 : 9 shape: the width ÷ the height is $16 \div 9 \approx 1.78$.

(a) A TV screen is 48 in wide and 27 in high. Is it similar to the 16 : 9 shape?

(b) An old TV screen is 40 cm wide and 30 cm high. Will a 16 : 9 movie fill the whole screen without being cut or stretched? Explain.

8. Ahmed drew a logo 20 cm wide and 25 cm high. The print shop made a sign 0.8 m wide and 1 m high. Is the sign similar to his drawing? If so, find the scale factor.

9. A 16 in by 20 in print is put in a frame that is 2 in wide on every side.

(a) Find the outside size of the frame.

(b) Is the framed print similar to the print alone? Show your work.

(c) Another frame's outside edge is similar to the print alone and has a short side of 20 in. How long should its long side be?

10. Parallelogram $EFGH$ has $EF = 8$ cm, $FG = 5$ cm and $\angle E = 60°$. Parallelogram $JKLM$ has $JK = 10$ cm, $KL = 6.25$ cm and $\angle J = 60°$.

(a) Find all the angles of both parallelograms.

(b) Check the side ratios. Are the parallelograms similar? If so, give the scale factor.

11. Maya wants a rectangle similar to a 10 cm by 4 cm rectangle. She adds 2 cm to each side and draws a 12 cm by 6 cm rectangle.

(a) Explain why Maya's rectangle is not similar.

(b) Her new rectangle must be 12 cm long. How wide should it be to be similar?

12. Regular polygons.

(a) Write *always*, *sometimes* or *never*.

Two regular pentagons are similar. _____ Two parallelograms are similar. _____

Two equilateral triangles are similar. _____ A square and a regular hexagon are similar. _____

(b) A patio uses paving stones shaped like regular hexagons. The small stones have 15 cm sides and the large stones have 24 cm sides. Are they similar? Find the size of each angle and the scale factor from small to large.

13. Three cereal boxes are shaped like rectangular prisms.
Box A: 20 cm by 7 cm by 30 cm Box B: 25 cm by 8.75 cm by 37.5 cm Box C: 24 cm by 8 cm by 36 cm
Which boxes are similar to Box A? Show your work.

14. Which pair of shapes is **always** similar?

A. two rectangles **B.** two rhombuses **C.** two regular hexagons **D.** two parallelograms

15. Which rectangle is similar to a 30 cm by 40 cm rectangle?

A. 35 cm by 45 cm **B.** 45 cm by 60 cm **C.** 40 cm by 50 cm **D.** 60 cm by 70 cm

16. A square has 4 cm sides. A rhombus has 6 cm sides and angles of 80° and 100°. Why are they **not** similar?

A. The side ratios are not all the same.

B. The corresponding angles are not equal.

C. They have a different number of sides.

D. The rhombus is bigger than the square.

17. Numerical Response Quadrilateral $ABCD$ has sides 16 cm, 20 cm, 12 cm and 24 cm. Quadrilateral $EFGH$ is similar to it, with corresponding sides 12 cm, 15 cm, 9 cm and 18 cm. The scale factor from $ABCD$ to $EFGH$, to the nearest hundredth, is _____.

18. Numerical Response A photo is 12 cm by 18 cm. A similar enlargement has a short side of 20 cm. The length of its long side, in centimetres, is _____.

Lesson 3: Drawing Enlargements and Reductions

Assignment

1. Is each copy an enlargement, a reduction or the same size?

k	(a) 2	(b) 0.8	(c) 1	(d) $\frac{3}{2}$	(e) $\frac{1}{4}$	(f) 1.05
Copy						

2. A triangle has sides of 5 cm, 8 cm and 12 cm, and one angle of 48°. Complete the table for each similar copy.

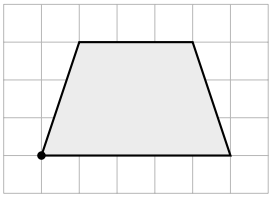
k	**5 cm side**	**8 cm side**	**12 cm side**	**The** 48° **angle**
2				
0.5				
1.5				
0.2				

3. A rectangle is 6 cm by 10 cm.

(a) Leon adds 4 cm to each side to get a 10 cm by 14 cm rectangle. Is it similar to the first one? Divide to show why or why not.

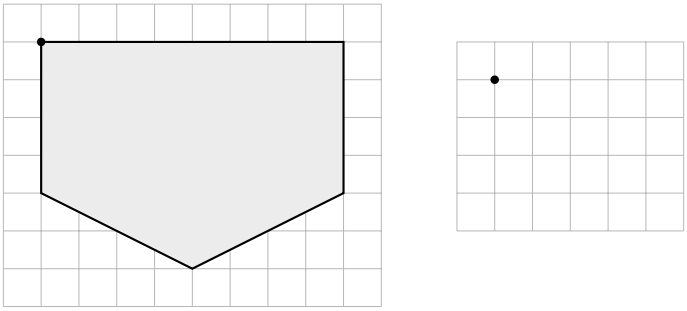
(b) Find the size of a similar rectangle whose long side is 15 cm.

4. Draw an enlargement of the trapezoid with $k = 2$. Start at the dot.

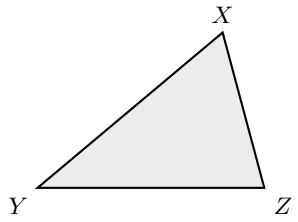




5. A hockey team's crest is a shield. Draw a reduction of the shield with $k = \frac{1}{2}$. Start at the dot.



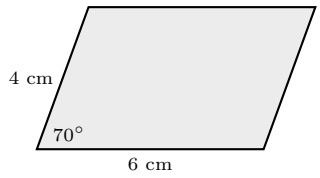
6. (a) Measure side YZ, $\angle Y$ and $\angle Z$ of $\triangle XYZ$.



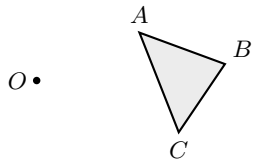
(b) Use a ruler and protractor to draw an enlargement with $k = 2$ in the space.

(c) Measure $\angle X$ and $\angle X'$. Are they equal?

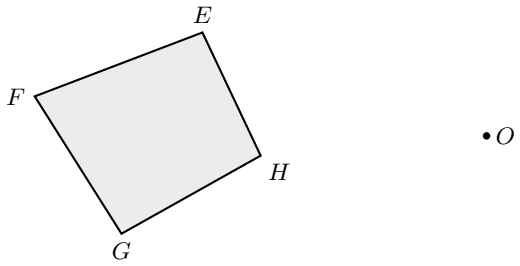
7. A sketch of a parallelogram is shown (not to scale). Draw a reduction of it with $k = 0.75$. Label your side lengths and angles.



8. Draw an enlargement of $\triangle ABC$ with centre O and $k = 2$. Write each distance from O and each new distance.



9. Draw a reduction of quadrilateral $EFGH$ with centre O and $k = \frac{1}{2}$. Write each distance from O and each new distance.



10. Amir uses the projection method with $k = 3$. Corner A is 2.5 cm from O. He marks A' on the ray, 7.5 cm *past* A. Is he right? Explain, and say how far from O his point really is.

11. Kalani draws a plan of a deck. On her drawing, the deck is 6 cm by 4.5 cm. The builder uses a scale factor of 80 to build it.

(a) What are the real length and width of the deck, in centimetres and in metres?

(b) The steps on the drawing are 1.2 cm wide. How wide are the real steps?

12. Owen has a 5 in by 7 in photo of his dog.

(a) What size is an enlargement with $k = 2$?

(b) Is an 11 in by 14 in print similar to the photo? Explain.

(c) He wants the largest similar print that fits in a frame with an 11 in by 14 in opening. Find k and the size of the print.

13. Sam was asked to draw a reduction of quadrilateral $PQRS$ with $k = 0.5$. Here are his measurements.

	PQ	QR	RS	SP	$\angle P$
Original	8 cm	6 cm	5 cm	9.4 cm	80°
Sam's copy	4 cm	3 cm	3.5 cm	4.7 cm	40°

Find his two mistakes and say what each measurement should be.

14. A rectangle is 3 cm by 5 cm. Which rectangle is an enlargement of it with $k = 4$?

A. 7 cm by 9 cm **B.** 12 cm by 20 cm **C.** 12 cm by 5 cm **D.** 0.75 cm by 1.25 cm

15. A triangle has a 35° angle. It is enlarged with $k = 3$. The matching angle in the enlargement is

A. 11.7° **B.** 35° **C.** 38° **D.** 105°

16. In the projection method, corner A is 6 cm from the centre O. For a reduction with $k = \frac{1}{3}$, how far from O is A'?

A. 2 cm **B.** 3 cm **C.** 9 cm **D.** 18 cm

17. Numerical Response A photo on a phone screen is 6.4 cm wide. It is printed as a poster with a scale factor of 12.5. The width of the poster, in centimetres, is _____ cm.

18. Numerical Response A sign shop has a logo drawing that is 8 cm wide. The finished sign must be 1.4 m wide, and similar to the drawing. The scale factor, to the nearest tenth, is _____.

Lesson 4: Scale Diagrams: Maps and Floor Plans

Assignment

1. Write each scale as a ratio. Then say if it is a reduction or an enlargement.

	Scale in words	**As a ratio**	**Reduction or enlargement?**
(a)	1 cm represents 4 m		
(b)	1 cm represents 10 km		
(c)	1 cm represents 25 cm		
(d)	1 in represents 2 ft		
(e)	5 cm represents 1 mm		

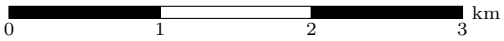
2. Write each ratio scale in words, using a sensible real unit (m or km).

(a) 1 : 300 _____

(b) 1 : 75 000 _____

(c) 1 : 5 000 000 _____

3. A trail map has this bar scale. Use a ruler.



(a) How long is one block? Write the scale in words, as "1 cm represents _____".

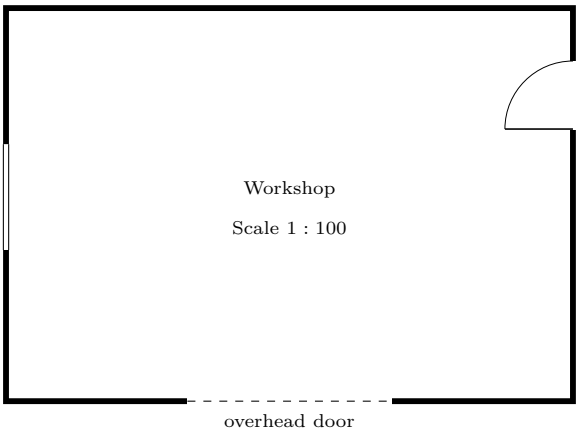
(b) Write the scale as a ratio.

(c) A lake on this map is 4.6 cm across. How wide is the real lake?

4. A cabin plan uses the scale 1 cm represents 2 m. Complete the table.

Part	**Plan (cm)**	**Real (m)**	**Work**
Cabin length	4.8		
Deck length	2.25		
Front door	0.45		
Cabin width		5.5	
Window		1.2	

5. This plan of a workshop is drawn to scale at 1 : 100. Use a ruler.



(a) Write the scale in words.

(b) Measure the plan. Find the real length and width of the workshop.

(c) How wide is the real overhead door?

(d) The floor will get an epoxy coating that costs $18.50 per square metre. Find the real floor area and the cost before tax.

(e) Find the total cost with 5% GST.

6. A road map of Alberta has a scale of 1 : 1 500 000.

(a) How many kilometres does 1 cm on the map represent?

(b) Calgary and Banff are 7 cm apart on the map. About how far apart are they really (straight line)?

(c) Red Deer and Edmonton are about 145 km apart (straight line). How far apart are they on this map, to the nearest tenth of a centimetre?

7. A repair manual shows a watch gear at a scale of 5 : 1.

(a) Is the drawing bigger or smaller than the real gear?

(b) The gear in the drawing is 4.2 cm across. How wide is the real gear, in millimetres?

(c) A real pin in the watch is 3 mm long. How long should it be in the drawing?

8. The plan for a backyard deck uses the scale 1 in represents 4 ft.

(a) The deck is 16 ft long and 12 ft wide. Find its length and width on the plan.

(b) The stairs are $3\frac{1}{2}$ ft wide. How wide are they on the plan, as a fraction of an inch?

(c) A railing is $2\frac{3}{4}$ in long on the plan. How long is the real railing?

9. Models are built to scale.

(a) An HO-scale (1 : 87) model locomotive is 24.5 cm long. How long is the real locomotive, to the nearest tenth of a metre?

(b) An old prairie grain elevator is 25 m tall. How tall is an HO model of it, to the nearest tenth of a centimetre?

(c) A model sailboat is built at a scale of 3 : 100. The model is 27 cm long. How long is the real boat, in metres?

(d) The real mast of that boat is 12 m tall. How tall is the model's mast?

10. Tomas wants to draw his backyard, 24 m long and 15 m wide, in a space 14 cm long and 10 cm wide.

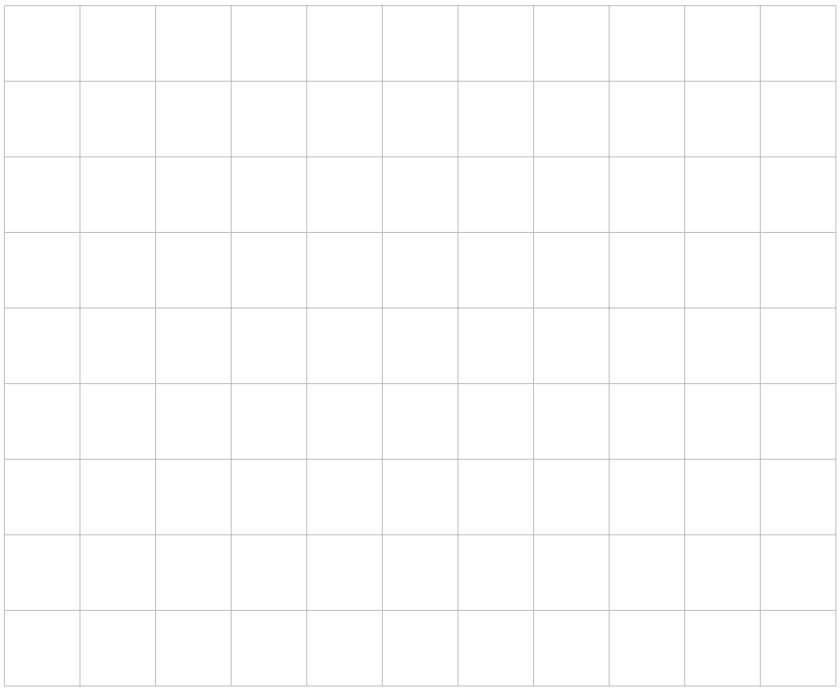
(a) Choose an easy scale so the yard fits. Show your work.

(b) Find the length and width of the yard on his drawing.

(c) Why would "1 cm represents 1.5 m" not work? Why is "1 cm represents 10 m" a poor choice?

11. Draw a floor plan of a kitchen using the scale 1 cm represents 0.4 m. Fill in the plan lengths first.
The kitchen is 4.0 m long and 3.2 m wide. It has a 0.8 m door near a corner of one long wall, a counter 0.6 m deep along the whole opposite long wall, and an island 1.6 m by 0.8 m in the middle.

Item	**Plan (cm)**
Length	
Width	
Door	
Counter depth	
Island	

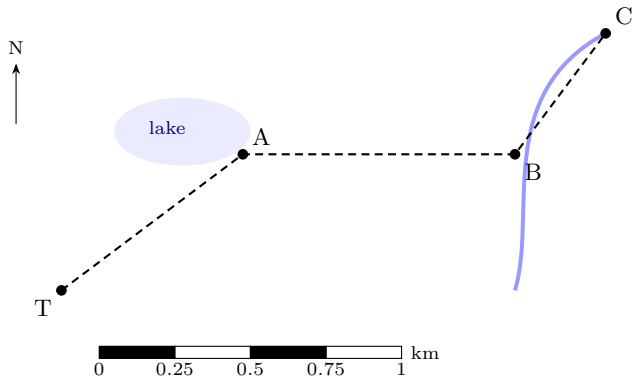


12. On a $1 : 150$ floor plan, a wall is 6.4 cm long. Kai says, "The real wall is $6.4 \times 150 = 960$ m long."

(a) How can you tell right away that Kai's answer is wrong?

(b) Find his mistake and the correct length.

13. This trail map is drawn to scale. The trail goes from the trailhead T past a lake (A) and a bridge (B) to a waterfall (C). Use a ruler.



(a) Write the scale in words.

(b) Measure each part of the trail. Find the real length of the trail from T to C.

(c) Find the straight-line distance from T to C. Why is it shorter than the trail?

(d) Rhys hikes from T to C and back along the trail at about 3 km/h. About how long does the hike take, in hours and minutes?

14. On a map with a scale of 1 : 25 000, two lookouts are 4 cm apart. How far apart are they really?

A. 1 km **B.** 10 km **C.** 100 m **D.** 6250 m

15. A model truck is built at a scale of 1 : 24. Which statement is true?

A. Every length on the model is $\frac{1}{24}$ of the real length.

B. Every real length is $\frac{1}{24}$ of the model length.

C. The model is 24 cm long.

D. Every angle on the model is $\frac{1}{24}$ of the real angle.

16. Numerical Response On a 1 : 75 floor plan, a room is 5.6 cm long. The real length of the room, in metres, is _____.

17. Numerical Response On a map, 1 cm represents 2.5 km. A bike route is 17 km long. On the map, the route is _____ cm long.

Lesson 5: Similar Triangles

Assignment

1. Find the missing angle in each triangle. Then write **yes** or **no**: are the two triangles similar?

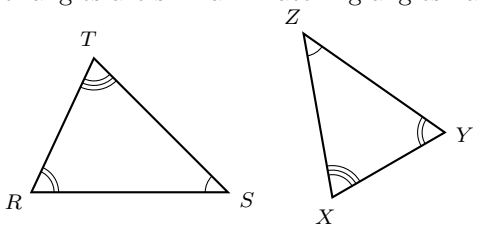
	Triangle 1 has	**Missing angle**	**Triangle 2 has**	**Missing angle**	**Similar?**
(a)	52° and 71°		57° and 52°		
(b)	90° and 38°		90° and 52°		
(c)	80° and 45°		80° and 50°		
(d)	110° and 25°		45° and 25°		

2. $\triangle GHJ \sim \triangle KLM$. Fill in the blanks.

(a) $\angle G = \angle$_____ (b) $\angle H = \angle$_____ (c) $\angle J = \angle$_____

(d) $\dfrac{KL}{GH} = \dfrac{LM}{____} = \dfrac{____}{GJ}$ (e) Side HJ matches side _____.

3. The two triangles are similar. Matching angles have the same number of arcs.



(a) Complete: $\triangle RST \sim \triangle$_____

(b) Which side matches RS? _____

(c) Which side matches ST? _____

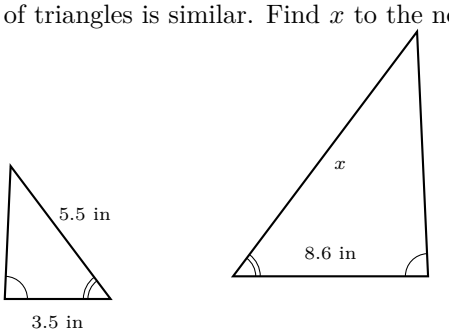
4. $\triangle ABC \sim \triangle DEF$. Matching sides are in the same column.

$\triangle ABC$	$AB = 4$ cm	$BC = 7$ cm	$AC = 9$ cm
$\triangle DEF$	$DE = 10$ cm	$EF =$	$DF =$

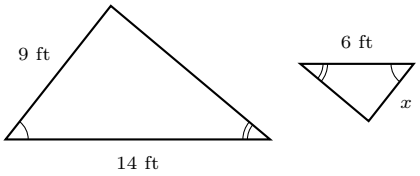
(a) Find the scale factor k from $\triangle ABC$ to $\triangle DEF$. Is $\triangle DEF$ an enlargement or a reduction?

(b) Find EF and DF. Write them in the table.

5. Each pair of triangles is similar. Find x to the nearest tenth.



(a)



(b)

6. $\triangle JKL \sim \triangle PQR$. Mia knows that $JK = 9$ cm, $KL = 12$ cm and $PR = 10$ cm.
(a) Mia says she cannot find any of the missing sides. Is she right? Explain.

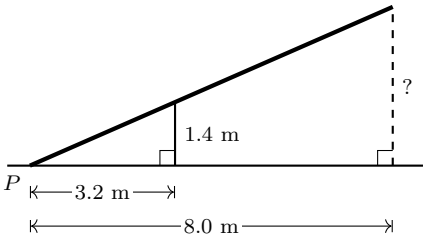
(b) Name **one** more length that would let her find all the missing sides.

7. Write **always**, **sometimes** or **never**. Explain each answer, or give an example.
(a) Two equilateral triangles (all sides equal, all angles 60°) are similar. _____

(b) Two right triangles are similar. _____

(c) Two isosceles triangles with a 50° angle at the top (between the equal sides) are similar. _____

8. A conveyor belt rises steadily from point P on the floor. A vertical support post is 3.2 m from P along the floor, and it is 1.4 m tall. The top end of the belt is 8.0 m from P along the floor.

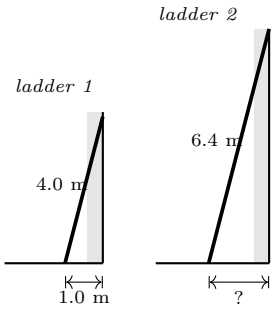


(a) Explain why the small and large right triangles are similar.

(b) How high above the floor is the top end of the belt?

9. Two ladders lean against a wall at the **same angle** with the level ground. Ladder 1 is 4.0 m long, and its foot is 1.0 m from the wall. Ladder 2 is 6.4 m long.

(a) Why are the two right triangles similar?

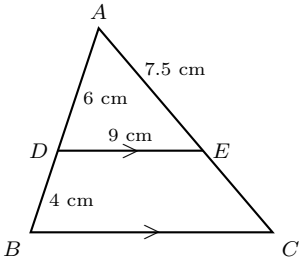


(b) How far from the wall is the foot of ladder 2?

(c) Ladder 1 reaches 3.9 m up the wall. About how high does ladder 2 reach? Round to the nearest tenth of a metre.

10. In $\triangle ABC$, $DE \parallel BC$. $AD = 6$ cm, $DB = 4$ cm, $DE = 9$ cm and $AE = 7.5$ cm.

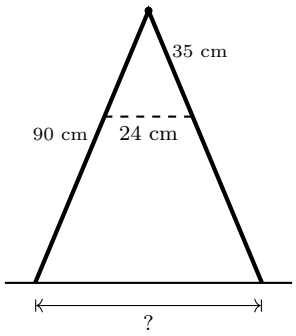
(a) Explain why $\triangle ADE \sim \triangle ABC$.



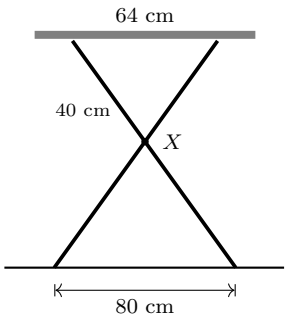
(b) Find BC.

(c) Find EC.

11. A sandwich-board sign is shaped like the letter A. The two legs are each 90 cm long, from the hinge at the top to the floor. A chain joins the legs 35 cm down from the hinge, and it is pulled tight, parallel to the floor. The chain is 24 cm long. How far apart are the feet of the sign, to the nearest tenth of a centimetre?



12. The legs of a folding sawhorse cross at point X. The top bar is parallel to the floor. The tops of the legs are 64 cm apart and the feet are 80 cm apart. Each leg measures 40 cm from its top to X.



(a) Give two reasons why the top triangle and the bottom triangle are similar.

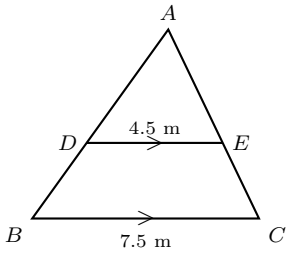
(b) How long is each leg from X to the floor?

13. In $\triangle ABC$, $DE \parallel BC$ (D on AB, E on AC). $AD = 8$ cm, $DB = 4$ cm and $DE = 10$ cm. Here is Jordan's work to find BC:

$$\frac{BC}{10} = \frac{4}{8} \qquad BC = 10 \times 4 \div 8 = 5 \text{ cm}$$

Find Jordan's mistake. Then find the correct length of BC.

14. In $\triangle ABC$, $DE \parallel BC$. $DE = 4.5$ m, $BC = 7.5$ m and the whole side $AB = 12$ m. Find AD and DB.



15. $\triangle KLM \sim \triangle RST$. Which ratio is equal to $\frac{KL}{RS}$?

A. $\frac{LM}{RT}$ **B.** $\frac{LM}{ST}$ **C.** $\frac{KM}{ST}$ **D.** $\frac{RS}{KL}$

16. Two angles of a triangle are 40° and 75°. A second triangle must be similar to it if two of its angles are

A. 40° and 60° **B.** 75° and 65° **C.** 40° and 105° **D.** 115° and 40°

17. In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. $AD = 3$ cm, $DB = 6$ cm and $DE = 4$ cm. What is the length of BC?

A. 8 cm **B.** 12 cm **C.** 2 cm **D.** 7 cm

18. Numerical Response $\triangle ABC \sim \triangle DEF$. $AB = 7.5$ cm, $DE = 12$ cm and $BC = 5.2$ cm. The length of EF, to the nearest tenth of a centimetre, is _____ cm.

Lesson 6: Indirect Measurement: Shadows and Mirrors

Assignment

1. Fill in the blanks.

(a) The sun's rays are _____, so at the same moment they hit nearby objects at the same angle.

(b) A flagpole on level ground makes a _____ angle with its shadow.

(c) With a mirror, the angle in is _____ the angle out.

(d) In the mirror method, use your _____ height, not your full height.

2. Explain why the triangle made by a person and her shadow is similar to the triangle made by a tree and its shadow, when both shadows are measured at the same time.

3. In each row, the two shadows were measured at the same time. Find the missing length. Show your proportion in the space below the table.

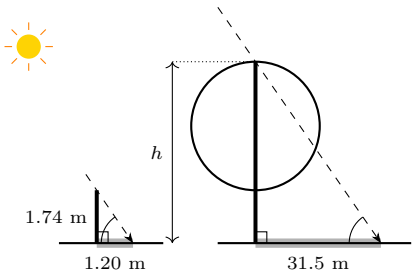
	Stick height	**Stick shadow**	**Object height**	**Object shadow**
(a)	1.0 m	0.8 m		6.4 m
(b)	2.0 m	3.0 m		15.6 m
(c)	1.2 m	0.9 m		18.0 m
(d)	1.5 m	2.5 m	6.0 m	

4. At noon in June, a 1.50 m tall child casts a shadow only 0.50 m long. At the same time, a light pole casts a shadow 2.9 m long.

(a) How many times as long as her shadow is the child?

(b) Use your answer to (a) to find the height of the light pole.

5. At a summer fair, Brooke is 1.74 m tall and casts a 1.20 m shadow. At the same moment, the Ferris wheel casts a shadow 31.5 m long.

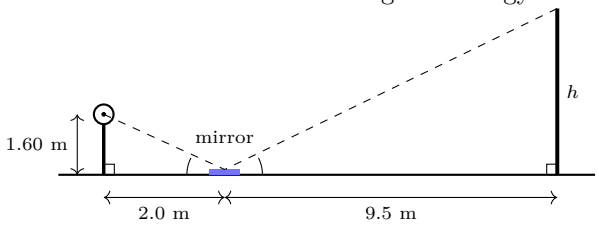


(a) Estimate the height of the Ferris wheel.

(b) Find the height of the Ferris wheel to the nearest tenth of a metre. Is it close to your estimate?

6. A girl who is 5 ft 3 in tall casts a shadow 7 ft long. At the same time, a flagpole casts a shadow 36 ft long. How tall is the flagpole? (Hint: change 5 ft 3 in to feet first.)

7. Hannah uses a mirror to find the height of the gym wall. Find the height h.



8. The rim of a basketball hoop is 3.05 m above the floor. Leo puts a mirror on the floor, 4.2 m from the point straight below the rim. His eyes are 1.45 m above the floor. How far from the mirror must he stand to see the rim in the mirror? Round to the nearest tenth of a metre.

9. Grace's eyes are 1.54 m above the ground. She stands 1.4 m from a mirror, and the mirror is 12.6 m from a building. She writes:

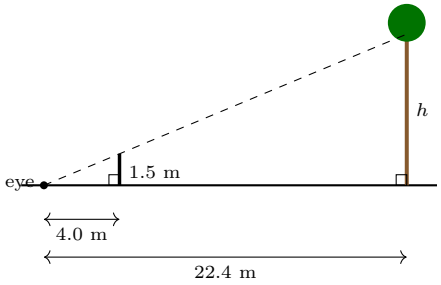
$$\frac{h}{14.0} = \frac{1.54}{1.4}, \text{ so } h = 15.4 \text{ m}.$$

(a) What mistake did Grace make?

(b) Find the correct height, to the nearest tenth of a metre.

10. Maddox holds a ruler upright 65 cm from his eye. A radio tower 250 m away covers 13 cm of the ruler, from its base to its top. How tall is the tower?

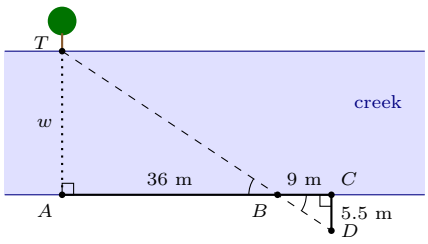
11. Ella lies on the ground with her eye at ground level. She pushes a 1.5 m stake into the ground 4.0 m from her eye. She moves until the top of the stake lines up with the top of a tree. Her eye is then 22.4 m from the tree.



(a) Why are the two triangles similar?

(b) How tall is the tree?

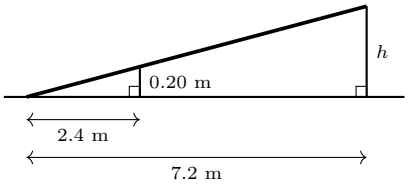
12. A creek runs through a coulee. To find its width w, Marcus sets up the stake method as shown: $AB = 36$ m, $BC = 9$ m and $CD = 5.5$ m.



(a) Write the proportion.

(b) How wide is the creek?

13. A wheelchair ramp rises at a steady slope. A support post stands 2.4 m along the ground from the bottom of the ramp, and it is 0.20 m tall. The top of the ramp is 7.2 m along the ground from the bottom. How high is the top of the ramp?



14. Sam measured the shadow of a 1.0 m stick: 1.5 m. The school's shadow at the same moment was 18 m. Sam says the school is 27 m tall.

(a) Without calculating, explain why 27 m can't be right.

(b) Find the correct height.

15. It is a cloudy day and you want the height of a tree. Which method from this lesson can you use, and which can't you use? Explain.

16. A person 1.7 m tall casts a 2.4 m shadow. At the same time, a tree casts a 14 m shadow. Which proportion gives the tree's height h?

A. $\frac{h}{14} = \frac{1.7}{2.4}$ **B.** $\frac{h}{14} = \frac{2.4}{1.7}$ **C.** $\frac{h}{1.7} = \frac{2.4}{14}$ **D.** $\frac{h}{14 + 2.4} = \frac{1.7}{2.4}$

17. Rosa is 1.65 m tall and her eyes are 1.50 m above the ground. She stands 3.0 m from a mirror on the ground, and the mirror is 12.0 m from a tree. How tall is the tree?

A. 6.0 m **B.** 24.0 m **C.** 6.6 m **D.** 7.5 m

18. Numerical Response A 2.40 m post casts a shadow 1.60 m long. At the same time, a church steeple casts a shadow 19.0 m long. The height of the steeple, to the nearest tenth of a metre, is _____ m.

19. Numerical Response A water tower is known to be 30 m tall. Tyler holds a ruler upright 60 cm from his eye, and the tower covers 4.5 cm of the ruler. His distance from the tower, to the nearest metre, is _____ m.

Lesson 7: Unit 6 Review

Assignment

This review is set up like the unit test. Show your work. Write each proportion before you solve it. Diagrams are not to scale unless the question says so.

Part A: Written Response

1. (a) Find the scale factor k (new ÷ original). Then write enlargement, reduction or congruent.

Original length	**New length**	***k***	**Type**
5 cm	15 cm		
12 cm	9 cm		
8 m	8 m		
2.5 in	6 in		

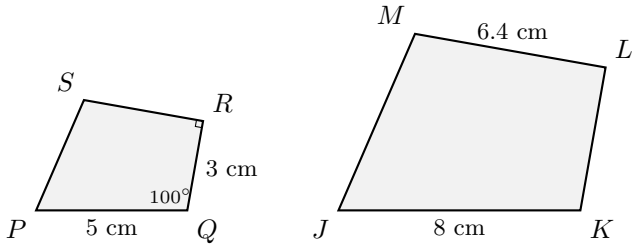
(b) Fill in the blanks.

$k = 4$, original 3.5 cm, new _____

$k = 0.6$, original 45 cm, new _____

$k = 2.5$, new 20 cm, original _____

2. $PQRS \sim JKLM$.



(a) Which side of $JKLM$ matches QR? Which angle matches $\angle S$?

(b) Find the scale factor from $PQRS$ to $JKLM$.

(c) Find KL.

(d) Find RS.

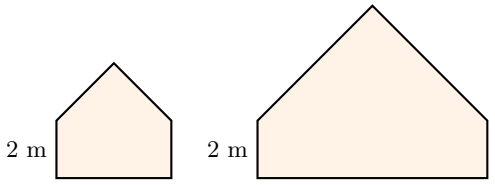
(e) Find $\angle K$ and $\angle L$.

3. Canada's flag is twice as long as it is wide. A school has three flags or banners: flag A is 75 cm by 150 cm, flag B is 1.8 m by 3.6 m, and banner C is 50 cm by 75 cm. All the corners are right angles.

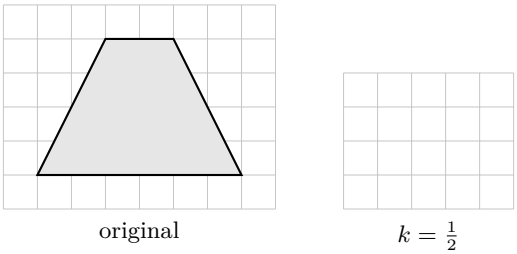
(a) Are flag A and flag B similar? Show your work.

(b) Are flag A and banner C similar? Show your work.

4. Rae says, "These two houses are similar, because every angle in house 1 matches an angle in house 2." Both houses have angles of 90°, 135°, 90°, 135° and 90°. Is Rae right? Explain.



5. Draw a reduction of the shape with a scale factor of $\frac{1}{2}$ on the grid on the right.



6. A sign maker enlarges a logo that is 40 cm by 25 cm, using $k = 3$. Jae says the new logo is 120 cm by 105 cm. "I made the long side 80 cm bigger, so I made the short side 80 cm bigger too."

(a) Explain Jae's mistake and find the correct size.

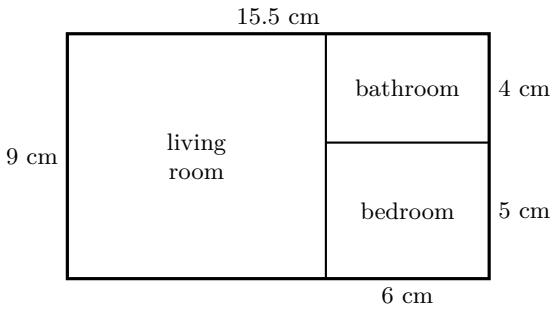
(b) The corners of the small logo are right angles. What are the corners of the big logo?

(c) How many times as much paint will the big logo need as the small one? (Compare the areas.)

7. Fill in the table. Each row shows the same scale three ways.

In words	As a ratio
1 cm represents 5 m	
	1 : 75 000
1 cm represents 2 km	

8. Here is a floor plan of a small cabin, drawn at a scale of 1 : 40. The measurements are the lengths on the plan.



(a) Write the scale in words, using metres.

(b) Find the real length and width of the bedroom.

(c) Find the real length and width of the whole cabin.

(d) A couch is 1.8 m long. How long should it be on the plan?

9. On a road map with a scale of 1 : 500 000, the drive from a farm to the nearest town measures 4.2 cm along one road and then 3.6 cm along another.

(a) What does 1 cm on the map represent, in kilometres?

(b) How long is the drive?

(c) At an average of 90 km/h, how many minutes does the drive take?

10. Decide if each pair of triangles is similar. If it is, name the triangles in matching order.

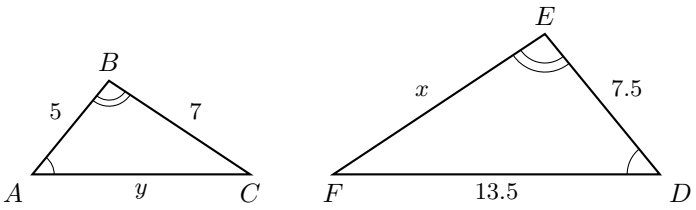
(a) $\triangle ABC$: $\angle A = 52°$, $\angle B = 61°$. $\triangle DEF$: $\angle D = 61°$, $\angle E = 67°$.

(b) $\triangle GHI$: $\angle G = 48°$, $\angle H = 70°$. $\triangle JKL$: $\angle J = 48°$, $\angle K = 72°$.

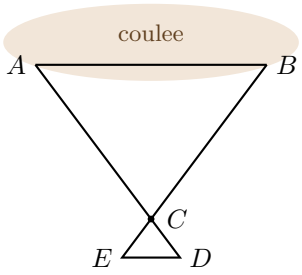
(c) Two right triangles each have a 35° angle. Explain why they must be similar.

11. $\triangle ABC \sim \triangle DEF$. The equal angles are marked. Find x and y.

Lengths are in centimetres.



12. To find the distance AB across a coulee near Drumheller, Kiran puts stakes at C, D and E so that A, C, D are in a line, B, C, E are in a line, and $DE \parallel AB$. She measures $CA = 40$ m, $CD = 10$ m and $DE = 7.5$ m.



(a) Explain why $\triangle CAB \sim \triangle CDE$.

(b) Find the distance across the coulee, AB.

13. At noon, a fence post 1.5 m tall casts a shadow 1.2 m long. At the same time, a church steeple casts a shadow 22 m long.

(a) How tall is the steeple?

(b) Why must both shadows be measured at the same time of day?

14. To find the height of a climbing wall in a gym, Lee puts a mirror on the floor 6 m from the wall. She stands 1.2 m from the mirror and sees the top of the wall in it. Her eyes are 1.7 m above the floor. How tall is the wall?

Part B: Multiple Choice

15. A photo 20 cm wide is reduced to 8 cm wide. The scale factor is

A. 2.5 **B.** 0.4 **C.** 12 **D.** 0.6

16. Which pair of figures is **always** similar?

A. two rectangles

B. two rhombuses

C. two regular hexagons

D. two isosceles triangles

17. $ABCD \sim EFGH$. The side that matches CD is

A. EF **B.** FG **C.** GH **D.** HE

18. A map scale of $1 : 250\,000$ means that 1 cm on the map represents

A. 250 m **B.** 2.5 km **C.** 25 km **D.** 250 km

19. A design is enlarged with $k = 2$. An angle of $70°$ in the design becomes

A. $140°$ **B.** $35°$ **C.** $72°$ **D.** $70°$

20. In $\triangle PQR$, $\angle P = 38°$ and $\angle Q = 90°$. Which triangle **must** be similar to $\triangle PQR$?

A. a right triangle with a $48°$ angle

B. a triangle with angles of $38°$ and $100°$

C. a right triangle with a $52°$ angle

D. a right triangle with two equal acute angles

21. A child 1.5 m tall casts a 2 m shadow. At the same time, a tree casts a 12 m shadow. The tree is

A. 16 m **B.** 9 m **C.** 13.5 m **D.** 10.5 m

22. On a floor plan with a scale of $1 : 25$, a wall 3 m long is drawn

A. 0.12 cm **B.** 0.75 cm **C.** 12 cm **D.** 75 m

Part C: Numerical Response

23. Numerical Response A 6 cm by 9 cm rectangle is enlarged so that its short side is 15 cm. The long side of the enlargement is _____ cm.

24. Numerical Response On a map, 1 cm represents 1.5 km. Two towns are 8.4 cm apart on the map. The real distance is _____ km.

25. Numerical Response Sam's eyes are 1.6 m above the ground. He sees the top of a building in a mirror on the ground. He is 2.5 m from the mirror, and the mirror is 20 m from the building. The building is _____ m tall (to the nearest tenth).

26. Numerical Response A picture 5 in wide is projected onto a wall, where it is 7 ft 6 in wide. The scale factor of the projection is _____.
